

when $\epsilon > 1$ [24], which makes the problem become trivial. Therefore they ignored these cases.

Recall that the values of rainbow (k, l) -connectivity may be different for distinct edge-colorings. In [21], Dellamonica, Magnant and Martin derived that a random k -coloring of a sufficiently large complete graph has asymptotically optimal rainbow (k, l) -connectivity (see Theorems 3.3 and 3.4). They obtained an *explicit* edge-coloring. By explicit, we mean that they gave a polynomial time algorithm to compute such an edge-coloring.

Recently, the computational complexity of rainbow vertex-connection numbers has been determined by Chen, Li and Shi [18].

Motivated by the proofs of Theorems 6.1 and 6.3, they derived two corresponding results to the rainbow vertex-connection.

Theorem 6.7 [18] *Given a graph G , deciding if $rvc(G) = 2$ is NP-Complete. In particular, computing $rvc(G)$ is NP-Hard.*

Theorem 6.8 [18] *The following problem is NP-Complete: given a vertex-colored graph G , check whether the given coloring makes G rainbow vertex-connected.*

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